PROBLEM 44

The Little Prince Throwing Seeds

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Angular Momentum + Vis-Viva

The little Prince is on his spherical asteroid, which has radius R and mass M, and does not rotate. While he sits at the equator, he throws a baobab seed with an initial velocity of magnitude v_0 and angle $\gamma = 60^\circ$ with respect to the local horizontal. Knowing that it reaches an apocenter of $r = 3R$ before falling back, determine v_0 as a function of the parameters of the problem and the universal gravitational constant G.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a seed of mass $m \ll M$ launched from the surface ($r_0 = R$) with speed v_0 at $\gamma = 60^\circ$ above the local horizontal; it reaches the apocenter $r_a = 3R$.
- **Hypotheses:** spherically symmetric, non-rotating asteroid, so the gravitational force is central and conservative with $\mu = GM$; vacuum (no drag); inertial frame S_0 centred on the asteroid.
- **Constraints:** none (the seed is a free particle once it leaves the surface), so $\mathrm{CDOF} = 3$. The central force makes the motion planar, so $\mathrm{CDOF} = 2$ with the coordinates (r, θ) in the orbital plane.
- **Initial state:** $r = R$, radial component $v_{r0} = v_0 \sin\gamma = \frac{\sqrt{3}}{2} v_0$ and transverse component $v_{\theta 0} = v_0 \cos\gamma = \frac{1}{2} v_0$. At the apocenter $\dot{r} = 0$.

Symbol	Meaning	SI unit
M, R	mass and radius of the asteroid	kg, m
G	universal gravitational constant, 6.6743×10^{-11}	$\mathrm{m^3/(kg\ s^2)}$
$\mu = GM$	gravitational parameter	$\mathrm{m^3/s^2}$

Symbol	Meaning	SI unit
γ	launch angle above the local horizontal (given: 60°)	rad
r_a	apocenter radius (given: $3R$)	m
v_0	launch speed (unknown)	m/s

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with O at the centre of the asteroid.
- **Polar basis in the orbital plane.** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by the true anomaly θ about $\mathbf{k}$ (along $\mathbf{H}_O$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

- $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$, so by Poisson $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$.

Derivation

3.1 Kinematics

Pedagogical Justification: the product rule and Poisson's formula (as in Problem 42, 3.1) give the position, velocity and acceleration in the polar basis. The velocity is what the conservation laws need, and the acceleration is needed to write Newton's second law in 3.3.

$$\mathbf{r} = r\mathbf{e}_r, \qquad \mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a} = \left(\ddot r - r\dot\theta^2\right)\mathbf{e}_r + \left(r\ddot\theta + 2\dot r\dot\theta\right)\mathbf{e}_\theta$$

$$h = \|\mathbf{r}\times\mathbf{v}\| = r^2\dot\theta = r\,v_\theta$$

- At launch, $\dot{r} = v_0 \sin\gamma$ and $r\dot{\theta} = v_0 \cos\gamma$. At the apocenter $\dot{r} = 0$, so the velocity is purely transverse: $v_a = r_a \dot{\theta}_a$.

3.2 Forces

- **Applied force:** the gravitation of the asteroid, $\mathbf{F} = -\dfrac{GMm}{r^2}\mathbf{e}_r$. It is central and conservative.
- **Constraint and inertia forces:** none (free particle, inertial frame).

> **Pedagogical Justification:** the potential energy per unit mass follows from Barrow's rule with the reference at infinity:

$$V_0(r) = -\int_{\infty}^{r} \left(-\frac{GM}{s^2}\right) ds = GM\left[-\frac{1}{s}\right]_{\infty}^{r} = GM\left[-\frac{1}{r} - \lim_{s\to\infty}\left(-\frac{1}{s}\right)\right] = -\frac{GM}{r}$$

3.3 Equations of Motion

> **Pedagogical Justification:** the unknowns are the orbit variables $r(t)$ and $\theta(t)$. Instead of integrating Newton's second law in time, we use its two first integrals, which follow from it and give the speed directly at two points (launch and apocenter), exactly what the question needs.

$$m\left(\ddot{r} - r\dot{\theta}^2\right) = -\frac{GMm}{r^2}, \qquad m\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) = 0$$

- **Angular momentum.** The transverse equation times r/m is $\frac{d}{dt}\left(r^2\dot{\theta}\right) = 0$, so $h = r^2\dot{\theta}$ is constant (the force is central).
- **Energy.** The force is conservative, so $\xi = \frac{1}{2}v^2 - GM/r$ is constant. Both laws hold for the same particle between launch and apocenter.
- **Unknowns and equations:** v_0 and v_a are unknown, and there are two equations (h and ξ conservation), so the problem is closed.

3.4 Integration and Answers

Conservation of specific angular momentum

> **Pedagogical Justification:** evaluate h at launch and at the apocenter, where $v_a = r_a \dot{\theta}_a$.

$$h = R\,(v_0 \cos\gamma) = \frac{1}{2} R v_0, \qquad h = r_a v_a = 3R v_a \implies v_a = \frac{h}{3R} = \frac{\frac{1}{2} R v_0}{3R} = \frac{v_0}{6}$$

Conservation of specific energy

> **Pedagogical Justification:** the energy takes the same value at launch and at the apocenter.

$$\frac{1}{2} v_0^2 - \frac{GM}{R} = \frac{1}{2}\left(\frac{v_0}{6}\right)^2 - \frac{GM}{3R} = \frac{v_0^2}{72} - \frac{GM}{3R}$$

Solving for v_0

$$\left(\frac{1}{2} - \frac{1}{72}\right) v_0^2 = \left(1 - \frac{1}{3}\right) \frac{GM}{R} \implies \frac{35}{72} v_0^2 = \frac{2}{3} \frac{GM}{R} \implies v_0^2 = \frac{2}{3} \cdot \frac{72}{35} \frac{GM}{R} = \frac{48}{35} \frac{GM}{R}$$

$$\boxed{v_0 = \sqrt{\frac{48}{35} \frac{GM}{R}}}$$

Interpretation & Checks

- **Bound orbit:** the escape speed at the surface is $v_e = \sqrt{2GM/R}$, i.e. $v_e^2 = \frac{70}{35} GM/R$. Since $\frac{48}{35} < 2$, $v_0 < v_e$ and the orbit is an ellipse: the seed falls back.
- **Orbital parameters:** $h^2 = \frac{1}{4} R^2 v_0^2 = \frac{12}{35} GMR$ and the semi-latus rectum $p = h^2/\mu = \frac{12}{35} R$. From $r_a = p/(1-e) = 3R$: $1 - e = \frac{4}{35}$ and $e = \frac{31}{35} \approx 0.8857$. The periapsis of the full conic is $r_p = p/(1+e) = \frac{2}{11} R < R$, inside the asteroid.

- **Physical arc.** On the orbit $r = p/(1 + e\cos\theta)$ the seed leaves the surface where $1 + e\cos\theta_0 = p/R$, i.e. $\cos\theta_0 = -\frac{23}{31}$ with $\sin\theta_0 > 0$ (the seed is moving away from the centre, $\dot r_0 > 0$), so $\theta_0 \approx 137.9^\circ$. It passes the apocenter at $\theta = \pi$ and lands where $r = R$ again, at $\theta = 2\pi - \theta_0 \approx 222.1^\circ$. The part of the conic that contains the periapsis ($\theta = 0$) is inside the asteroid and is not travelled: the seed describes about 84° of orbit and falls back.
- **Independent check with the Binet solution.** The orbit equation of Problem 42 for $\alpha = 2$ is $u'' + u = \mu/h^2$, with the first integral $\left(u - \mu/h^2\right)^2 + (u')^2 = \left(\mu e/h^2\right)^2$ that defines the eccentricity. The launch data give $u_0 = 1/R$ and $u_0' = -\dot r_0/h = -\dfrac{(\sqrt{3}/2)v_0}{(1/2)Rv_0} = -\dfrac{\sqrt{3}}{R}$, so

$$\frac{\mu e}{h^2} = \sqrt{\left(\frac{1}{R} - \frac{35}{12R}\right)^2 + \frac{3}{R^2}} = \sqrt{\frac{529}{144R^2} + \frac{432}{144R^2}} = \frac{31}{12R} \implies e = \frac{31}{12R}\cdot\frac{12R}{35} = \frac{31}{35}$$

- This eccentricity, obtained from the launch conditions alone, reproduces the prescribed apocenter: $r_a = \dfrac{p}{1-e} = \dfrac{12R/35}{4/35} = 3R$. The value of v_0 found from energy and angular momentum is therefore consistent with a second, independent formulation.
- **Limiting cases** of the general result. Repeating the algebra with r_a and γ as symbols ($h = Rv_0\cos\gamma$, $v_a = (R/r_a)v_0\cos\gamma$):

$$v_0^2 = \frac{2\mu\left(\frac{1}{R} - \frac{1}{r_a}\right)}{1 - \left(\frac{R}{r_a}\right)^2\cos^2\gamma}$$

For $r_a = 3R$ and $\gamma = 60^\circ$ the denominator is $1 - \frac{1}{9}\cdot\frac{1}{4} = \frac{35}{36}$ and $v_0^2 = \frac{4}{3}\cdot\frac{36}{35}\,\mu/R = \frac{48}{35}\,\mu/R$. As $r_a \to \infty$, $v_0^2 \to 2\mu/R$, the escape speed. For $\gamma = 90^\circ$ (vertical launch) $v_0^2 = 2\mu\,(1/R - 1/r_a)$, the energy balance of a radial flight. For $\gamma = 0$ and $r_a \to R$ the quotient tends to μ/R, the circular speed.

- **Dimensions:** $[v_0] = \sqrt{[G][M]/[R]} = \sqrt{\dfrac{\mathrm{m}^3}{\mathrm{kg\,s}^2} \cdot \dfrac{\mathrm{kg}}{\mathrm{m}}} = \mathrm{m/s}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 44 in its solution keys, so the result is verified by independent derivation: sympy solves the energy equation and returns $v_0^2 = \frac{48}{35} GM/R$, and the eccentricity obtained from the launch data reproduces the prescribed apocenter $r_a = 3R$.
- **Numerical check** (Python, DOP853 integration of $\ddot{\mathbf{r}} = -\mu \mathbf{r}/r^3$ with $\mu = R = 1$ and $v_0 = \sqrt{48/35}$, $\gamma = 60^\circ$): the maximum radius reached is $2.99998R$ (sampling resolution of the output) and the speed there is $0.16668\, v_0$, in agreement with $r_a = 3R$ and $v_a = v_0/6$.

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